

Deva Anand

(413) 315-1715 | devaanand@umass.edu | devaanand.com | linkedin.com/in/devaaa | github.com/Deva-1903

Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, Java, SQL, JavaScript, TypeScript, C++, Bash

Data & Performance

Apache Spark / PySpark, Spark SQL, Spark UI, ETL pipelines, distributed data processing, Linux profiling, performance optimization

Backend & Reliability

Node.js, Express, FastAPI, REST APIs, integration testing, code review, authentication hardening, rate limiting, multi-tenant SaaS

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

Databases & Tools

PostgreSQL, MongoDB, Redis, SQLite, Agile / Scrum, pytest

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned backend reliability for **multi-tenant production systems** (Node.js, MongoDB) serving **8+ UK clients** and **10,000+ monthly submissions**; monitored production logs, troubleshoot and shipped fixes for incidents, and added **integration tests** to maintain code quality and protect against regressions.
- Contributed to **clearing technical debt**: routed tech-debt PRs, code migrations, and security-finding triage through a structured (AI-assisted) review workflow, and remediated **20+ VAPT findings** across tenant isolation and backend access paths.
- Architected a full-stack **data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label and curate data for production models; cut manual work by **100+ hours/month** and improved throughput by **60%**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety and long-term maintainability through targeted refactoring.

Projects

Spark ETL Performance Optimization - Big-Data Pipeline

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI, SQL | ~1.4B-row NYC TLC dataset

- Optimized a Spark **ETL pipeline** over a ~**1.4B-row** dataset, replacing shuffle joins with broadcast joins (minutes to ~0.05s), enabling Adaptive Query Execution and skew-join handling, and tuning shuffle partitions, schema-aware batched reads, and partitioning, cutting end-to-end runtime ~**25%** (17–19 min to 14.3 min).
- Reduced analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x** via partition and column pruning and filter pushdown; profiled stage-level timings with **Spark UI**, added a 6-rule **data-quality filter** (retaining 95.26% of records), and documented a repeatable optimization playbook.

AgenticSearch - Tested, Instrumented Data Pipeline

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage extraction pipeline that plans typed retrieval, reranks pages, extracts entities with deterministic parsers before LLM fallback, and verifies outputs, returning cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, provider routing and fallback, and pipeline instrumentation for review-friendly debuggability and troubleshooting.

cf-ai-research-scout - Fault-Tolerant Workflow System

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers, Durable Objects, WorkflowEntrypoint

- Designed durable workflow boundaries so multi-step tasks survive process restarts and tool failures, with state persisted server-side without an external database, an exercise in fault-tolerant system design.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.